Solution

(i) $\left(50000000\right)^4 = \left(5.0\times10^7\right)^4$

$= \left(5.0\right)^4 \times \left(10^7\right)^4$

$= 625.0\times10^{28}$

$= 6.25\times10^2\times10^{28}$

$= 6.25\times10^{30}$

(ii) $\left(0.00000005\right)^3 = \left(5.0\times10^{-8}\right)^3$

$= \left(5.0\right)^3 \times \left(10^{-8}\right)^3$

$= \left(125.0\right)\times\left(10\right)^{-24}$

$= 1.25\times10^2\times10^{-24}$

$= 1.25\times10^{-22}$

(iii) $\left(300000\right)^3 \times \left(2000\right)^4$

$= \left(3.0\times10^5\right)^3 \times \left(2.0\times10^3\right)^4$

$= \left(3.0\right)^3 \times \left(10^5\right)^3 \times \left(2.0\right)^4 \times \left(10^3\right)^4$

$= \left(27.0\right)\times\left(10^{15}\right)\times\left(16.0\right)\times\left(10^{12}\right)$

$= \left(2.7\times10^1\right)\times\left(10^{15}\right)\times\left(1.6\times10^1\right)\times\left(10^{12}\right)$

$= 2.7\times1.6\times10^1\times10^{15}\times10^1\times10^{12}$

$= 4.32\times10^{1+15+1+12} \quad = 4.32\times10^{29}$

(iv) $\left(4000000\right)^3 \div \left(0.00002\right)^4$

$= \left(4.0\times10^6\right)^3 \div \left(2.0\times10^{-5}\right)^4$

$= \left(4.0\right)^3 \times \left(10^6\right)^3 \div \left(2.0\right)^4 \times \left(10^{-5}\right)^4$

$= \dfrac{64.0\times10^{18}}{16.0\times10^{-20}}$

$= 4\times10^{18}\times10^{+20}$

$= 4.0\times10^{38}$

Thinking Corner

1. Write two numbers in scientific notation whose product is 2. 83104.
2. Write two numbers in scientific notation whose quotient is 2. 83104.

Exercise 2.8

1. Represent the following numbers in the scientific notation:
   (i) 569430000000 (ii) 2000.57
   (iii) 0.0000006000 (iv) 0.0009000002

2. Write the following numbers in decimal form:
   (i) 3.459×10^6 (ii) 5.678×10^4
   (iii) 1.00005×10^{-5} (iv) 2.530009×10^{-7}

3. Represent the following numbers in scientific notation:
   (i) $\left(300000\right)^2 \times \left(20000\right)^4$ (ii) $\left(0.000001\right)^{11} \div \left(0.005\right)^3$
   (iii) $\left\{\left(0.00003\right)^6 \times \left(0.00005\right)^4\right\} \div \left\{\left(0.009\right)^3 \times \left(0.05\right)^2\right\}$

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